

A Note on Unrestricted Chow–Liu Tree Selection for Mixed Discrete–Continuous Data

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Abstract

The Chow–Liu algorithm summarises the dependency structure of mixed discrete–continuous data as a tree of strongest pairwise associations. Earlier versions exclude trees in which a continuous variable connects two discrete variables, because such a path complicates the later density-fitting step. We show that this restriction is unnecessary for tree selection, which depends only on pairwise mutual-information values, well-defined for every combination of variable types. The restriction applies only to the later density-fitting step. We prove consistency of the unrestricted procedure and give finite-sample error bounds under a tail assumption on the pairwise mutual-information estimates. On synthetic mixed graphs and a synthetic gene-network benchmark, the unrestricted procedure recovers trees that the restricted procedure cannot. On the fully discrete Asia Bayesian benchmark, where Edwards’ restriction is inactive by construction, the pipeline attains the tree-approximation ceiling. A breast-cancer gene-expression dataset is included as an illustration, since no ground truth is available. The comparison with a discretisation-based estimator is regime-dependent. In the sample-size sweep, the discretisation estimator performs better at $n = 75$, the k -nearest-neighbour estimator performs better at $n = 1,500$, and the two are statistically indistinguishable at $n \in \{100, 200, 500, 1,000\}$.

Keywords: Chow–Liu algorithm; mixed graphical models; mutual information; structure learning; conditional Gaussian model.

1 Introduction

Consider a study that records a panel of p variables for n individuals. Some variables are categorical, such as genotypes at single nucleotide polymorphism (SNP) sites, and some are continuous, such as gene-expression levels or disease biomarkers. A common first task is to summarise the pairwise dependencies among these variables. The Chow–Liu algorithm [3] answers a tractable version of this task. It returns the tree-structured graph whose edges together capture as much of the joint dependency as possible, measured by the sum of pairwise mutual informations (MI). The optimal tree is the maximum-weight spanning tree (MST) of the complete graph whose edge weights are pairwise MI values. Equivalently, the chosen tree minimises the Kullback–Leibler (KL) divergence to the true joint distribution among all tree-structured approximations.

This mixed-type setting arises naturally in genetics [18, 20] and in epidemiology where disease status accompanies continuous biomarkers. The conditional Gaussian (CG) model [14, 16] is the standard joint distribution for such data. A modelling issue arises in the CG framework. If the tree contains a path D_1-G-D_2 in which G is continuous and D_1, D_2 are discrete, then the tree factorisation requires the local conditional $P(D_2 | G)$, which is not part of the standard CG parametrisation [25]. To avoid this difficulty, Edwards et al. [6] imposed a structural restriction on the tree search. The restriction forbids continuous nodes from separating discrete nodes. It guarantees a fully parametrised CG model on the learned tree, but it also prevents the algorithm from recovering $D-G-D$ Markov paths that arise naturally in mixed-type data. Suzuki [22] subsequently evaded the restriction in practice by discretising continuous variables with a nested mesh and applying a minimum description length (MDL) penalty to the discretised mutual information. The restriction is sometimes treated as part of the mixed-type Chow–Liu procedure, although it constrains only the later CG parametrisation step.

We observe that the restriction belongs to the later density-fitting step, not to the Chow–Liu objective itself. The objective is a sum of pairwise mutual informations, each of which is symmetric in its two arguments. The objective therefore does not require any directional factor such as $P(D | G)$. Under standard regularity conditions, the unrestricted procedure recovers the population Chow–Liu tree as the sample size grows. We state this consistency result in Proposition 1 and a forest analogue under a

shrinking threshold in Proposition 2. A finite-sample bound (Corollary 1) and a tighter per-swap bound (Corollary 2) follow. Our contribution is the clarification, its empirical validation, and the supporting finite-sample analysis. The proof techniques are standard.

Contributions.

1. *Separation of structure selection from CG parametrisation.* The Chow–Liu objective is a sum of pairwise mutual informations. Each is a symmetric quantity that can be written using either $P(G | D)$ or $P(D | G)$ through the standard entropy decomposition. The objective therefore does not require modelling $P(D | G)$ for any pair. Edwards’ restriction applies only to the subsequent step of fitting a fully parametric CG density on the selected tree.
2. *An estimator-agnostic procedure with finite-sample bounds.* Algorithm 1 uses the Kraskov–Stögbauer–Grassberger (KSG) k -nearest-neighbour estimator for continuous–continuous pairs [12], the mixed-pair k -NN estimator of Gao et al. [7] for continuous–discrete pairs, and the empirical plug-in for discrete–discrete pairs. Proposition 1 establishes consistency whenever the pairwise mutual-information estimates are per-pair consistent. Under an additional sub-Gaussian deviation assumption, Corollaries 1 and 2 give finite-sample tree-recovery bounds. The per-swap bound in Corollary 2 uses the fact that exchanging one edge for another in a maximum spanning tree creates exactly one cycle, and bounds the probability of a single incorrect exchange. It improves the coarser tree-symmetric-difference bound in Corollary 1 by a factor of $(p - 1)^2$ in the exponent.
3. *Regime-dependent empirical comparison.* We compare the proposed pipeline (with the KSG MI estimator as input) against the Edwards-restricted MST baseline and against the MDL-discretisation MI estimator of Suzuki [22] over a range of graph sizes, effect sizes, conditional families, and sample sizes. The proposed pipeline outperforms the Edwards-restricted baseline across the tested graph-size and effect-size settings on graphs containing D - G - D paths. The comparison with the MDL-discretisation estimator is sample-size dependent. MDL discretisation performs better at $n = 75$ (where its bin selection acts as a form of model selection), the two estimators are statistically indistinguishable at $n \in \{100, 200, 500, 1,000\}$, and KSG performs better at $n = 1,500$.

Our results cover only the first step of the Chow–Liu workflow, selecting which edges form the tree, and the returned graph is undirected. A learned edge means that two variables are statistically associated; it carries no causal interpretation. A separate step, when needed, writes the joint distribution as an explicit parametric density on the chosen tree. The standard conditional-Gaussian density uses local factors of the form $P(G | D)$, but not $P(D | G)$. If the chosen tree contains a D - G - D path, this density-fitting step needs the factor $P(D | G)$, which lies outside the standard parametrisation. That step still requires either Edwards’ restriction, which forbids the path in the first place, or an alternative parametrisation of $P(D | G)$. We do not address this density-fitting step.

2 Background

2.1 Chow–Liu algorithm

A graphical model represents the joint distribution of p random variables $X_1, \dots, X_p$ as a graph whose edges encode pairwise dependencies. When the graph is constrained to be a tree, each variable is connected to the rest of the graph through a single path, and the joint distribution factorises into a product of pairwise marginals along the tree edges and univariate marginals at the nodes. The Chow–Liu algorithm seeks the tree T^* that best approximates the joint distribution P in Kullback–Leibler (KL) divergence:

$$T^* = \arg \max_{T \in \mathcal{T}} \sum_{(i,j) \in T} I(X_i; X_j), \quad (1)$$

where $\mathcal{T}$ is the set of all spanning trees on p vertices and $I(X_i; X_j)$ is the pairwise mutual information between X_i and X_j . The right-hand side of (1) is the total information captured by a tree, so the procedure selects the tree that captures as much pairwise dependency as possible. The optimisation reduces to Kruskal’s algorithm [13] applied to the complete graph weighted by the pairwise mutual informations. Kruskal’s algorithm adds edges in decreasing order of weight, skipping any edge that would form a cycle, until $p - 1$ edges have been added. The classical Chow–Liu algorithm therefore

returns a spanning tree even when some pairwise mutual informations are estimated as zero or near-zero. In the thresholded variant used below (Proposition 2), edges with estimated weight at or below a threshold τ_n are excluded before the Kruskal step is run, so the output may be a forest rather than a connected tree. Tree-structured graphical models remain a standard component of the probabilistic graphical models toolkit [10] because of this clean optimality property and the efficiency of Kruskal’s algorithm.

2.2 Mixed-type mutual information estimation

Mutual information is well-defined for any pair of random variables, but its estimation method depends on whether the variables are continuous, discrete, or mixed. For purely continuous pairs under a joint Gaussian assumption, the mutual information has the closed form $I(X; Y) = -\frac{1}{2} \log(1 - \rho^2)$, where ρ is the Pearson correlation coefficient [4]. For purely discrete pairs, the mutual information is estimated by the empirical plug-in, in which the joint and marginal probabilities are replaced by their sample frequencies. For mixed pairs, where one variable is continuous and the other discrete, neither of these forms applies. A distribution-free alternative is the k -nearest-neighbour estimator of Kraskov et al. [12], which builds on the Kozachenko and Leonenko [11] entropy estimator and was extended to mixed discrete–continuous variables with provably consistent convergence rates by Gao et al. [7]. The method uses distances between each sample point and its k -th nearest neighbour, and is consistent under standard regularity conditions on the underlying distribution. We use $k = 5$ as the default value throughout. For continuous–continuous pairs in which a joint Gaussian assumption is not appropriate, we use the Kraskov–Stögbauer–Grassberger (KSG) estimator [12], a k -nearest-neighbour estimator from the same paper that is designed for the two-continuous-variable case.

2.3 Edwards’ restriction

In the conditional Gaussian model, the tree factorisation takes the form $P(X_1, \dots, X_p) = \prod_i P(X_i \mid X_{\text{pa}(i)})$, where $\text{pa}(i)$ denotes the parent of node i in the tree rooted at any node. When each node is conditioned on a parent of the same type, the conditional distributions on the right-hand side are standard. The difficulty arises when the parent and child have different types. For a tree containing a path D_1-G-D_2 rooted at D_1 , the factorisation becomes $P = P(D_1) \cdot P(G \mid D_1) \cdot P(D_2 \mid G)$. The first two factors are standard in the conditional Gaussian model. $P(D_1)$ is a discrete distribution and $P(G \mid D_1)$ is the Gaussian conditional that defines the model. The third factor, $P(D_2 \mid G)$, is the conditional distribution of a discrete variable given a continuous one, which is not part of the standard CG parametrisation. Edwards et al. [6] avoided this difficulty by restricting Kruskal’s algorithm during tree selection. The modified Kruskal’s algorithm rejects any candidate edge whose addition would place a continuous node between two discrete components. The resulting tree always admits a full parametrisation within the standard CG model, but the procedure cannot recover any $D-G-D$ Markov chain, even when the data-generating distribution contains one. Figure 1 illustrates the configuration that the restriction forbids.

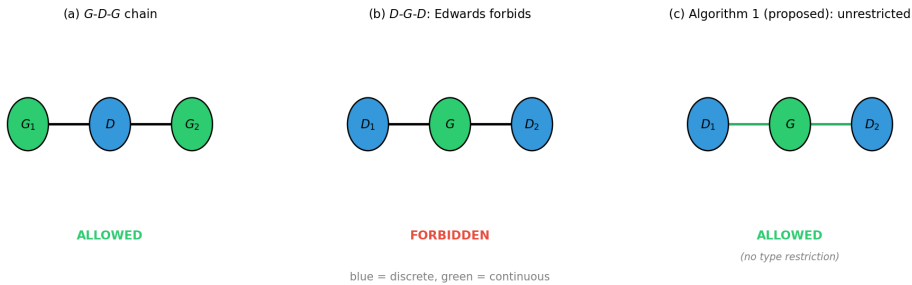


Figure 1: The $D-G-D$ configuration that Edwards’ restriction forbids by construction. A continuous variable G lies on the path between two discrete variables D_1 and D_2 , so the factorisation requires $P(D_2 \mid G)$, which is not part of the standard CG parametrisation.

2.4 Related work

Structure learning for graphical models is a well-developed area, with methods ranging from constraint-based to score-based and from parametric to nonparametric. Drton and Maathuis [5] provide a recent

overview of the field. Within mixed-type graphical modelling, Yang et al. [26] proposed ℓ_1 -regularised models in which the joint distribution is represented as a member of an exponential family. Their approach handles arbitrary graph structures rather than being restricted to trees, but it targets a parametric Markov random field and requires convex optimisation rather than a combinatorial tree search. On the mutual information side, Belghazi et al. [1] and Poole et al. [19] developed neural variational lower bounds on mutual information (the mutual-information neural estimator, MINE, and the noise-contrastive estimator, InfoNCE) used in representation learning. McAllester and Stratos [17] showed that any such variational bound has fundamental limitations in the distribution-free setting. Matching the true mutual information requires sample sizes that grow exponentially in the mutual information itself. The closest prior work to ours is Suzuki [22], who proposed an MDL-discretisation MI estimator and used it inside an unrestricted Chow–Liu procedure, thereby lifting Edwards’ restriction in practice via discretisation of the continuous variables. We discuss the MDL-discretisation procedure in detail and benchmark against it empirically in Section 4.7.

3 Theoretical statements

3.1 Consistency of the unrestricted Chow–Liu procedure

We refer to two quantities throughout. The first is the unrestricted Chow–Liu tree, obtained by maximising the total pairwise mutual information over all spanning trees. Proposition 1 concerns this quantity. The second is a thresholded maximum spanning forest, obtained by first discarding edges with weight at or below a threshold τ_n and then applying Kruskal’s algorithm to the remaining edges. Proposition 2 concerns this thresholded forest under additional separation assumptions. The zero-threshold rule used in many of our experiments returns a forest when some empirical mutual information estimates are zero or negative, but it is not a direct instance of Proposition 2.

Proposition 1 (Unrestricted Chow–Liu is well-defined for mixed types). *Let $X_1, \dots, X_p$ be random variables of arbitrary types. Assume*

(A1) *the pairwise mutual informations $I(X_i; X_j)$ are well-defined and finite for all pairs;*

(A2) *the maximum-weight spanning tree is unique;*

(A3) *for each pair, $\hat{I}_n(X_i; X_j) \xrightarrow{P} I(X_i; X_j)$ as $n \rightarrow \infty$.*

Then the unrestricted Chow–Liu algorithm recovers T^ with probability tending to 1, regardless of whether T^* contains D-G-D paths.*

The proof follows the standard M-estimation argument over the finite set of spanning trees [23]. Uniform convergence over $\mathcal{T}$ follows from (A3) applied to each of the $\binom{p}{2}$ pairs, and identification follows from (A2). The essential ingredient is the symmetry identity in equation (2), recorded as a remark after the proof, which makes explicit that the conditional $P(D \mid G)$ is not required to evaluate the Chow–Liu objective. The conditional may exist as a regular conditional distribution, but it is not a standard local component of the conditional Gaussian factorisation. Under an additional sub-Gaussian deviation assumption on the pairwise mutual information estimates with a common parameter $\sigma > 0$ (stated formally as $P(|\hat{I}_n - I| > t) \leq 2 \exp(-nt^2/(2\sigma^2))$ in Corollary 1), two finite-sample bounds follow. The first is a tree-symmetric-difference bound (Corollary 1); the second is a tighter per-swap bound (Corollary 2) that uses a per-edge cycle-completion argument for Kruskal’s algorithm and yields sample complexity $O(\sigma^2 \log p / \gamma_{\text{swap}}^2)$, where γ_{swap} is the smallest population mutual-information gap across all single-edge swaps in T^* (formal definition at (5) and positivity under (A2) in Corollary 2). Under (A2) the per-swap gap coincides with the tree-symmetric-difference gap at the population level (Remark after Corollary 2).

We do not establish the sub-Gaussian assumption itself. Empirical variance-scale diagnostics for the continuous–continuous KSG estimator and the mixed continuous–discrete k -NN MI estimator [7] are reported in Appendix B.

Before stating the forest analogue, we make three additional assumptions explicit. Let F^* denote the population thresholded maximum spanning forest, that is, the forest obtained from Kruskal’s algorithm on the population MI matrix after discarding all edges with $I(X_i; X_j) = 0$. The sample-level estimator uses a shrinking threshold $\tau_n > 0$ in place of the population zero-threshold rule, and the two thresholds agree only in the limit $\tau_n \rightarrow 0$. We assume:

- (A2') The population thresholded maximum spanning forest F^* is unique, equivalently the maximum spanning tree on each connected component of the support graph $\{(i, j) : I(X_i; X_j) > 0\}$ is unique.
- (A4) Every edge in F^* has mutual information at least $\delta > 0$.
- (A5) For every pair (i, j) whose population mutual information is zero, the estimator satisfies $P(|\hat{I}_n(X_i; X_j)| > \tau_n) \rightarrow 0$ as $n \rightarrow \infty$.

Proposition 2 (Forest consistency under shrinking-threshold rule). *Under (A1), (A3), (A2'), (A4), (A5), with $\tau_n \rightarrow 0$ and $\tau_n \sqrt{n} \rightarrow \infty$, the estimated forest $\hat{F}_n$ obtained from Kruskal's algorithm with threshold τ_n satisfies $P(\hat{F}_n = F^*) \rightarrow 1$.*

The main forest experiments use $\tau_n = 0$ to match Edwards' procedure, which uses no threshold. The shrinking-threshold condition of Proposition 2 ($\tau_n \rightarrow 0$, $\tau_n \sqrt{n} \rightarrow \infty$) is therefore not formally satisfied in those experiments. Section 4.6 reports a separate experiment that does satisfy the condition.

3.2 The unrestricted Chow–Liu algorithm (Algorithm 1)

Algorithm 1: Unrestricted mixed-type Chow–Liu algorithm

Input: Variables $X_1, \dots, X_p$ with types $t_i \in \{c, d\}$ (c = continuous, d = discrete); data $\{x^{(1)}, \dots, x^{(n)}\}$; threshold $\tau_n \geq 0$ (default $\tau_n = 0$); neighbour count $k \in \mathbb{N}$ for the k -NN estimators (default $k = 5$).

1. For each pair (i, j) with $i < j$, compute the pairwise mutual-information estimate according to the pair's type:
 - if $t_i = c$ and $t_j = c$: $\hat{w}_{ij} \leftarrow \hat{I}_{\text{KSG}}^{(k)}(X_i; X_j)$ [12];
 - if $t_i = d$ and $t_j = d$: $\hat{w}_{ij} \leftarrow \hat{I}_{\text{plug-in}}(X_i; X_j)$;
 - otherwise (one continuous and one discrete): $\hat{w}_{ij} \leftarrow \hat{I}_{k\text{-NN}}^{(k), \text{mix}}(X_i; X_j)$ [7].
2. Exclude every edge with $\hat{w}_{ij} \leq \tau_n$.
3. $\hat{F}_n \leftarrow \text{MaxSpanningForest}(\{\hat{w}_{ij}\})$, computed by Kruskal's algorithm with no type restriction.

Return: $\hat{F}_n$ (a spanning tree on p vertices when $\tau_n = 0$ and no edge is excluded, a strict forest otherwise).

Algorithm 1b (Gaussian variant). For computational efficiency on large graphs with many continuous–continuous pairs, the closed-form Gaussian MI $\hat{I}_{cc}(X; Y) = -\frac{1}{2} \log(1 - \hat{\rho}^2)$ can replace the KSG estimator. We refer to this variant as Algorithm 1b. It is faster but is consistent only under joint Gaussianity. We use Algorithm 1b only in the breast-cancer illustration (Section 4.10).

Algorithm 1a (Edwards-restricted baseline). Throughout the experiments we refer to Algorithm 1a as the Edwards-restricted version of Algorithm 1, in which Kruskal's loop rejects any candidate edge that would place a continuous node X_c on the unique path between two existing discrete components. Algorithm 1a uses the same MI estimators and the same Kruskal ordering as Algorithm 1. Only the type-check that decides whether to accept a candidate edge differs. The empirical comparisons in Section 4 use Algorithm 1a as the restricted baseline.

What Algorithm 1 provides relative to the MDL-discretisation procedure of Suzuki [22]. Both procedures lift Edwards' restriction and admit D - G - D paths. Algorithm 1 differs from the MDL-discretisation procedure in four respects. (i) It operates on the continuous data directly, with no information loss due to discretisation (by the data-processing inequality, $I(\tilde{G}; D) \leq I(G; D)$ for any binning $\tilde{G}$ of G , with equality only as the bin count tends to infinity). (ii) It has a single hyperparameter (k , default 5), rather than a bin scheme, a bin-count grid, and an MDL penalty constant. (iii) Proposition 1 applies to any consistent pairwise MI estimator, so Algorithm 1 is one instance among many, whereas the MDL-discretisation procedure is tied to its specific discretisation and penalty scheme. (iv) Proposition 1 gives a direct consistency statement for the Chow–Liu objective on the original continuous distribution, whereas the MDL framework gives a description-length justification on the discretised surrogate. The empirical comparison in Section 4.7 shows that these methodological differences translate into a sample-size-dependent practical recommendation.

3.3 Requirements on the mutual-information estimator

Proposition 1 requires only per-pair consistency (A3). The k -NN estimator [7, 12] satisfies (A3) under standard regularity conditions. The empirical plug-in for discrete pairs is consistent. The Gaussian formula is consistent only under joint Gaussianity. For variational mutual information estimators (MINE, InfoNCE), McAllester and Stratos [17] showed that achieving $o(1)$ bias requires sample complexity exponential in the true mutual information in the distribution-free setting. We therefore exclude these estimators from the default procedure.

4 Experiments

All experiments use fixed random seeds and run on a single CPU core. A single self-contained reproduction script (`260516_chow_liu.py`) regenerates every numerical result and every figure in the manuscript with the same publication-grade settings used here. The script requires only NumPy, SciPy, scikit-learn, pandas, matplotlib, and networkx. The full reproduction script completes in approximately four minutes on a single CPU core (laptop-grade hardware), and we report this figure only as an order-of-magnitude indication.

We report two standard graph-recovery metrics throughout. The F_1 score is the harmonic mean of edge precision (the fraction of recovered edges that are correct) and edge recall (the fraction of true edges that are recovered). The value $F_1 = 1$ corresponds to exact recovery. The structural Hamming distance (SHD) is the number of edge insertions and deletions needed to turn the recovered tree into the ground-truth tree; a lower SHD indicates better recovery.

4.1 Accuracy of the mutual-information estimators

We compare three estimators on four conditional distributions of $G \mid D = d$, with $n = 2,000$ samples and 15 trials per scenario. The true MI is computed by numerical integration. The conditional families are Gaussian (location-shifted), bimodal (mixture of two Gaussians), skewed (location-shifted χ_3^2), and heavy-tailed (location-shifted Student- t_3).

Table 1: MI estimation root mean squared error (RMSE), reported in units of 10^{-2} , on four conditional distributions ($n = 2,000$, 15 trials).

Scenario	True MI	ANOVA	Discret.	k -NN
Gaussian	0.2538	1.13	3.38	1.55
Bimodal	0.6053	36.80	8.44	2.15
Skewed	0.1612	10.80	5.30	1.79
Heavy-tail	0.1738	8.09	3.01	1.93

ANOVA is optimal on Gaussian data but heavily biased on Bimodal (RMSE about $17\times$ larger than that of k -NN) and on Skewed (RMSE about $6\times$ larger). The k -NN estimator is competitive across all four conditional families. We therefore use the k -NN estimator (KSG for continuous–continuous pairs and the mixed-pair variant of Gao et al. [7] for continuous–discrete pairs) in the empirical comparisons that follow, because the structure-learning experiments cover heterogeneous-conditional regimes in which ANOVA’s linearity assumption does not hold. Figure 2 visualises the estimator distributions across the four conditional families, and Figure 3 shows convergence of the three estimators with increasing sample size on Gaussian data, the one benchmark on which ANOVA’s assumption holds exactly.

4.2 D - G - D recovery (sanity check)

We test the simplest possible D - G - D configuration: $D_1 \sim \text{Uniform}\{0, 1, 2\}$, $G \sim \mathcal{N}(0, 1) + 1.5 D_1$, and D_2 is a binary indicator of whether G exceeds its median. The true Markov structure is D_1 - G - D_2 , a path that Edwards’ restriction forbids by construction. In 50 trials at $n = 1,500$ the unrestricted procedure recovers the path in 100% of trials, and the restricted procedure in 0%. The difference is representational rather than statistical: even with arbitrarily many samples, the restricted procedure cannot return this structure. Figure 4 contrasts the structures the two procedures can return.

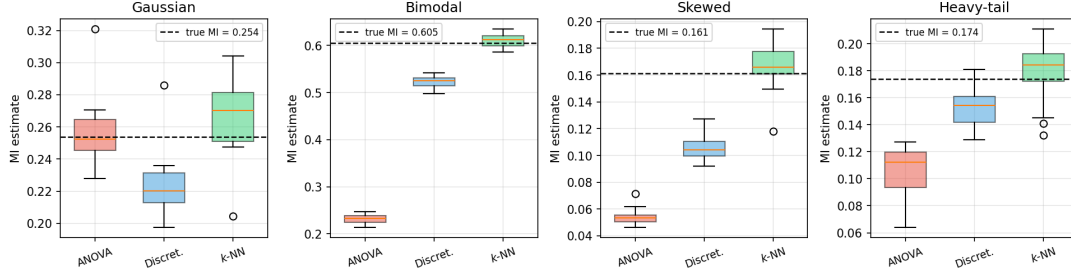


Figure 2: Mutual information estimation accuracy on four conditional distributions of $G \mid D = d$ (Gaussian, bimodal, skewed, heavy-tailed), each at $n = 2,000$ with 15 trials. The dashed line is the true MI computed by numerical integration. The k -NN estimator is competitive across all four families; ANOVA is severely biased on the non-Gaussian families.

4.3 Synthetic expression quantitative trait locus (eQTL) network

We consider an eight-variable expression quantitative trait locus (eQTL) network with 3 SNPs, 4 gene expressions, and 1 disease indicator, including the D - G - D Markov path $S_1 \rightarrow G_1 \rightarrow \text{Disease}$. We use $n = 1,500$ and 30 trials.

Table 2: Synthetic eQTL recovery, with structural Hamming distance (SHD) defined as the number of edge insertions and deletions required to transform the recovered tree into the ground-truth tree ($n = 1,500$, 30 trials).

Method	Precision	Recall	F_1	SHD
Alg. 1 (proposed) + k -NN (KSG)	0.714	1.000	0.833	2.00
Alg. 1a (Edwards-restr.) + k -NN (KSG)	0.571	0.800	0.667	4.00

Every trial returns the same edge set; the variance across trials is zero, so the recovery is deterministic at this signal level. The network and the procedure outputs are shown in Figure 5.

4.4 Random mixed graphs

We use a Prüfer-sequence random-mixed-tree generator with rejection sampling enforcing at least two D - G - D paths in every trial. Data are generated in breadth-first search (BFS) order from a randomly chosen root, using the parent-to-child rules described in Appendix A.1. We use Algorithm 1 (k -NN throughout) on every benchmark. Each table below draws an independent set of trials from this generator with its own random seed, so the same nominal configuration (for example, $p = 15$, $n = 1,500$, effect 0.6) gives slightly different mean F_1 across tables; the variation across tables is therefore a rough indicator of generator-level variability.

(a) Scaling with p . We use $n_c = \lfloor 8p/15 \rfloor$ continuous nodes, $n = 1,500$, 20 trials per row, and effect size 0.6.

Table 3: Scaling with p on random mixed graphs ($n = 1,500$, 20 trials, effect size = 0.6).

p	F_1 Alg. 1 (prop.)	F_1 Alg. 1a (Edw.-restr.)	SHD diff.	paired t	P
5	0.600 ± 0.166	0.438 ± 0.108	-1.3	+4.95	0.0001
8	0.679 ± 0.201	0.543 ± 0.147	-1.9	+5.15	0.0001
12	0.645 ± 0.118	0.573 ± 0.108	-1.6	+3.76	0.0013
15	0.739 ± 0.149	0.639 ± 0.136	-2.8	+4.17	0.0005
20	0.684 ± 0.131	0.637 ± 0.110	-1.8	+2.39	0.0273
30	0.728 ± 0.079	0.669 ± 0.078	-3.4	+4.01	0.0007

(b) Effect-size robustness ($p = 15$, $n = 1,500$, 20 trials).

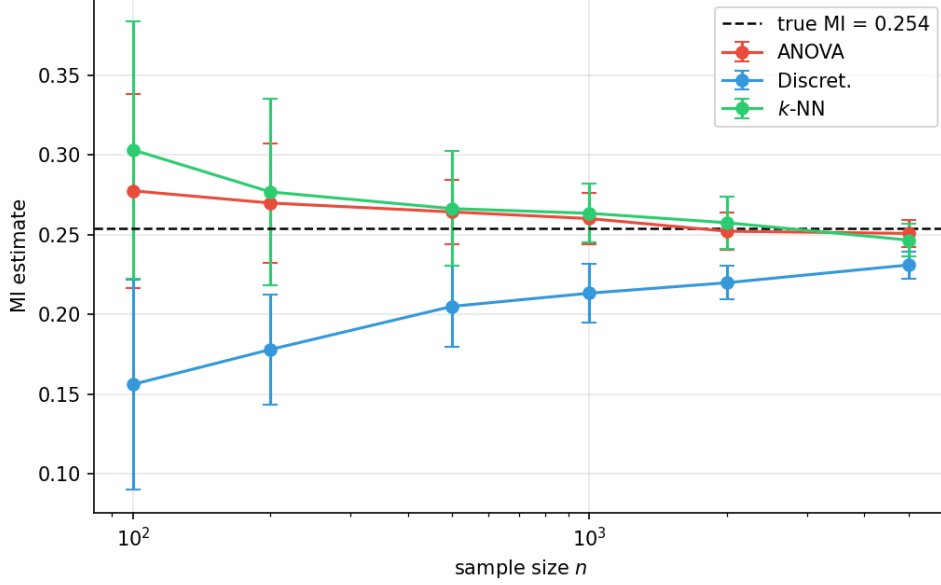


Figure 3: Convergence of the three mutual-information estimators with increasing sample size n on the Gaussian benchmark. The dashed line is the true MI. ANOVA has the smallest bias at every sample size, because its linearity assumption holds exactly on this benchmark. The k -nearest-neighbour estimator approaches the truth from above with slightly larger small-sample bias than ANOVA. The discretisation estimator has the largest small-sample downward bias and the slowest approach to the true value, but also approaches it as n grows. All three estimators are consistent on the Gaussian benchmark.

Table 4: Effect-size sweep at $p = 15$.

effect	F_1 Alg. 1 (prop.)	F_1 Alg. 1a (Edw.-restr.)	paired t	P
0.4	0.575 ± 0.127	0.536 ± 0.133	+2.60	0.0174
0.6	0.714 ± 0.093	0.650 ± 0.110	+4.16	0.0005
0.8	0.746 ± 0.129	0.646 ± 0.129	+5.72	< 0.0001
1.0	0.768 ± 0.115	0.675 ± 0.080	+4.95	0.0001
1.5	0.818 ± 0.114	0.689 ± 0.094	+9.00	< 0.0001
2.0	0.775 ± 0.116	0.661 ± 0.095	+4.77	0.0001

The paired advantage is significant across all effect sizes from 0.4 to 2.0. For interpretation, the true population mutual information of a Gaussian-shift edge at each effect size is reported in Table 5. The regime spans true MI from ≈ 0.05 nats (effect 0.4) to ≈ 0.62 nats (effect 2.0).

(c) **Heterogeneous-distribution data** ($p = 15$, $n = 1,500$, 20 trials, effect size = 0.6): each $c \mid d$ edge independently uses one of four conditional families, chosen uniformly at random. The families are Gaussian $\mathcal{N}(\text{effect} \cdot d, 1)$, bimodal, skewed (χ^2_3 -shift), and heavy-tailed (t_3 -shift).

Table 6 reports the result. Under heterogeneous conditional families, Algorithm 1 paired with the KSG estimator and Algorithm 1 paired with MDL discretisation yield indistinguishable mean F_1 ($P = 0.88$). The proposed pipeline significantly outperforms the Edwards-restricted baseline ($P = 0.0039$). The advantage of removing the type-pair restriction therefore holds under distributional heterogeneity; the choice between the two MI estimators shows no significant difference on this benchmark at $n = 1,500$.

4.5 Sample-size sweep on the D - G - D chain

We revisit the three-node D_1 - G - D_2 chain of Section 4.2 ($D_1 \sim \text{Uniform}\{0, 1, 2\}$, G Gaussian shifted by D_1 , D_2 a thresholded function of G) and vary the sample size. Chain transitivity gives the direct D_1 - D_2 pair a non-zero population mutual information that lies below the two true path edges. A noisy MI estimate could therefore swap the true edge G - D_2 for the spurious D_1 - D_2 edge during the Kruskal step, making this configuration a finite-sample stress test for the unrestricted procedure. It recovers the

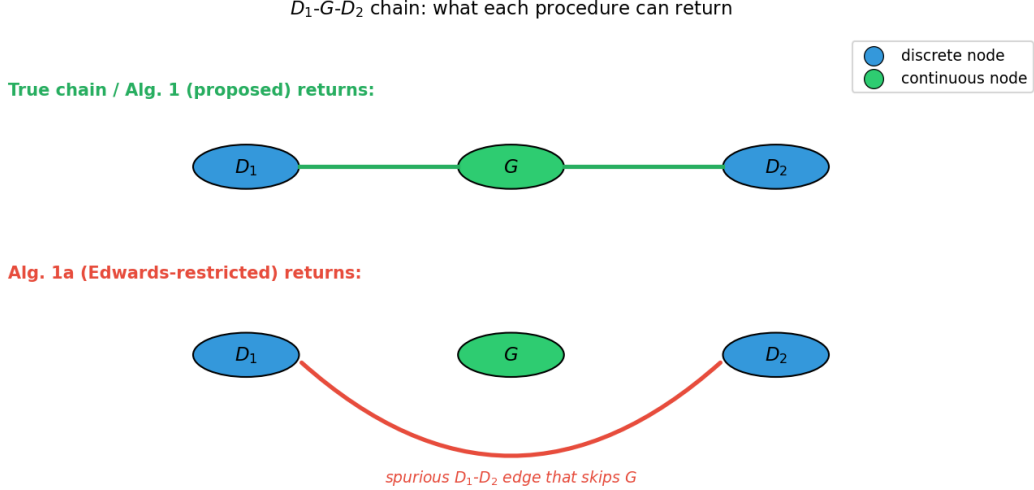


Figure 4: Structural comparison on a D_1 - G - D_2 chain. Top: Algorithm 1 returns the true Markov path through the continuous G . Bottom: Algorithm 1a, the Edwards-restricted baseline, cannot represent the path through G and instead returns the spurious D_1 - D_2 edge that skips G . This is a representational constraint, not a sample-size effect. Recovery counts: Algorithm 1 50/50 = 100%; Algorithm 1a 0/50 = 0%, by construction.

Table 5: Population mutual information of a Gaussian-shift edge with $K = 3$ discrete categories, unit conditional variance, at each effect size used in Table 4. Computed by numerical integration.

effect	true $I(G; D)$ (nats)
0.4	0.0507
0.6	0.1075
0.8	0.1772
1.0	0.2538
1.5	0.4472
2.0	0.6191

D - G - D chain in 100% of trials across $n \in \{500, 1000, 2000, 5000\}$, while the restricted procedure recovers 0% by construction.

4.6 Shrinking-threshold variant of Algorithm 1

We run Algorithm 1 with $\tau_n = c \log(n)/\sqrt{n}$ for $c \in \{0, 0.1, 0.5, 1.0\}$, on $p = 15$ random mixed graphs from the generator used in Section 4.4, with $n = 1,500$, 20 trials, and effect size 0.6. The setting $c = 0$ corresponds to the zero-threshold rule used elsewhere, while $c \in \{0.1, 0.5, 1.0\}$ satisfies the conditions of Proposition 2.

At $n = 1,500$, the threshold $c = 0.1$ excludes no true edges. In practice, at $n = 1,500$ the choice $c = 0.1$ yields F_1 and SHD identical to the zero-threshold rule (Table 7) while satisfying the consistency conditions of Proposition 2.

4.7 Sample-size sweep against the MDL-discretisation estimator

The MDL-discretisation MI estimator of Suzuki [22] discretises each continuous variable and uses the MDL-penalised empirical MI $\hat{I}_{\text{MDL}}(\tilde{G}; D) = \hat{I}(\tilde{G}; D) - (M - 1)(C - 1) \log(n)/(2n)$. We implement this estimator by sweeping bin counts $M \in \{2, 4, 8, 16, 32\}$ and taking the maximum penalised MI, and feed its output into the same unrestricted Kruskal step as Algorithm 1.

Synthetic eQTL benchmark ($n = 1,500$, 30 trials). Algorithm 1 paired with the MDL-discretisation MI estimator and Algorithm 1 paired with the KSG estimator produce identical results. Both yield

Synthetic eQTL network (red: $S_1 - G_1$ - Disease chain forbidden under Edwards)

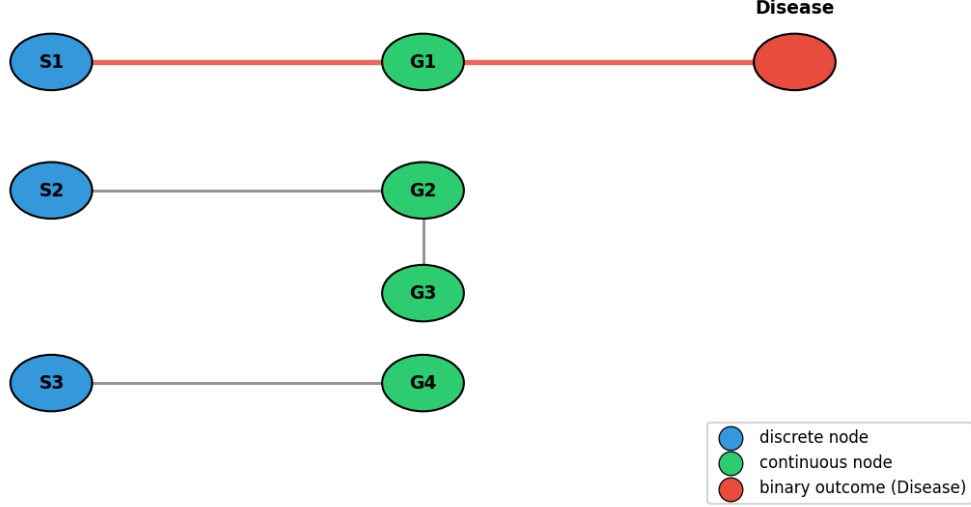


Figure 5: Synthetic eQTL network. Discrete nodes (blue): SNPs S_1, S_2, S_3 ; continuous nodes (green): gene expressions $G_1, \dots, G_4$; binary outcome (red): Disease. The red edges form the $D-G-D$ Markov path $S_1 \rightarrow G_1 \rightarrow \text{Disease}$ that Edwards’ restriction forbids by construction. F_1 recovery and SHD for each procedure are reported in Table 2.

Table 6: Heterogeneous-distribution experiment. The MDL-discretisation MI estimator [22] is added as a comparator; paired t -tests are against the proposed pipeline with the KSG estimator.

Method	F_1	paired t (P) vs Alg. 1 + KSG
Alg. 1 (proposed) + k -NN (KSG)	0.707 ± 0.098	reference
Alg. 1 (proposed) + MDL discr. MI	0.704 ± 0.116	+0.15 ($P = 0.8833$)
Alg. 1a (Edwards-restr.) + k -NN (KSG)	0.654 ± 0.099	+3.29 ($P = 0.0039$)

$F_1 = 0.833$, SHD = 2.0, and both recover the $S_1 \rightarrow G_1 \rightarrow \text{Disease}$ chain. The Edwards-restricted baseline yields $F_1 = 0.667$ and SHD = 4.0.

Random mixed graphs benchmark, sample-size sweep ($p = 15, 20$ trials per row, effect size = 0.6).

We report paired t -tests on F_1 over the same simulated graphs. The labels indicate which procedure has the larger mean F_1 when the test is significant at $\alpha = 0.05$; otherwise the entry is marked “n.s.” (not significant).

A qualitative regime crossover is visible between the two extremes. Denote the two pairings as Alg. 1 + MDL and Alg. 1 + KSG. At $n = 75$, Alg. 1 + MDL yields significantly higher F_1 ($P = 0.04$), because the MDL penalty adaptively favours coarser bins when samples are sparse, which acts as a form of model selection. At $n \in \{100, 200, 500\}$ the two estimators are statistically indistinguishable on F_1 ($P \in \{0.23, 0.19, 0.27\}$ respectively). At $n = 1,000$, Alg. 1 + KSG has slightly higher mean F_1 , but the paired comparison is not significant at $\alpha = 0.05$ ($P = 0.097$). At $n = 1,500$, Alg. 1 + KSG is significantly better ($P = 0.027$). The discretisation step in the MDL estimator introduces a finite-sample bias that grows relative to its variance benefit as n increases. (The data-processing inequality says only that $I(\hat{G}; D) \leq I(G; D)$ at the population level; at finite sample sizes, discretisation is a bias–variance trade-off, and the MDL penalty adaptively selects coarser bins to favour variance reduction under data sparsity.) At both ends of the sample-size range the unrestricted Algorithm 1 yields higher mean F_1 than the Edwards-restricted baseline.

Adaptive- k variant (exploratory). An exploratory adaptive- k check is reported in Appendix C; it narrows but does not eliminate the small-sample advantage of the MDL-discretisation estimator, and is

Table 7: Shrinking-threshold variant.

c	τ_n at $n = 1500$	F_1	SHD	Precision	Recall
0.0	0.000	0.679 ± 0.127	9.0 ± 3.6	0.679	0.679
0.1	0.019	0.679 ± 0.127	9.0 ± 3.6	0.679	0.679
0.5	0.094	0.562 ± 0.136	10.4 ± 3.0	0.677	0.482
1.0	0.189	0.466 ± 0.086	11.0 ± 1.8	0.731	0.343

Table 8: Sample-size sweep on $p = 15$ random graphs.

n	F_1 Alg. 1 + KSG	F_1 Alg. 1 + MDL discr.	F_1 Alg. 1a (Edw.-restr.) + KSG
75	0.529 ± 0.086	0.575 ± 0.107	0.521 ± 0.085
100	0.568 ± 0.136	0.542 ± 0.116	0.521 ± 0.132
200	0.625 ± 0.087	0.660 ± 0.091	0.586 ± 0.080
500	0.589 ± 0.133	0.619 ± 0.106	0.543 ± 0.120
1000	0.707 ± 0.093	0.664 ± 0.099	0.632 ± 0.079
1500	0.736 ± 0.122	0.686 ± 0.104	0.657 ± 0.086

not used in support of the main $n = 1,500$ comparison.

4.8 Mutual-information estimator ablation

To check whether the recovery improvement reported in Section 4.3 depends on the specific MI estimator, we compare three estimators (ANOVA, fixed-grid discretisation, and k -NN (KSG)) on the same eQTL benchmark at $n = 1,500$ over 20 trials. All three are used as inputs to Algorithm 1.

On this benchmark, all three MI estimators produce identical recovery. The structural advantage of removing the restriction (0.833 versus 0.667 in F_1) is independent of the choice. We recommend k -NN as the default.

4.9 Fully discrete sanity check: Asia network

We include the Asia Bayesian network [15] as a sanity check on a canonical fully discrete benchmark. Asia has 8 binary variables and 8 directed acyclic graph (DAG) skeleton edges. We use the 5000-sample dataset from the **gRbase** R package [9]. For $n \in \{500, 1,000, 2,500\}$, we draw 15 subsamples without replacement from the full 5000-sample dataset; the $n = 5,000$ row uses the full dataset directly. Asia is fully discrete, so Edwards’ type-pair restriction is inactive and all three procedures (Alg. 1 + KSG, Alg. 1a + KSG, Alg. 1 + MDL discretisation) produce identical trees by construction. The benchmark therefore serves to confirm that the proposed pipeline recovers a known canonical structure on real data, rather than to provide a comparison among methods. The Asia skeleton has 8 edges, but any spanning tree on 8 nodes contains only 7 edges, so a Chow–Liu tree cannot recover the full skeleton. The maximum F_1 attainable by a tree approximation of an 8-node, 8-edge graph occurs when all 7 tree edges are true positives: precision = $7/7 = 1$, recall = $7/8$, and $F_1 = \frac{2 \cdot 1 \cdot (7/8)}{1 + 7/8} = 14/15 \approx 0.933$. Table 11 reports exactly this value at $n = 5,000$, so the procedure attains the tree-approximation ceiling.

At $n = 5,000$, the procedure recovers 7 of the 8 DAG skeleton edges in every subsample. This matches the maximum number of skeleton edges that any spanning tree on 8 nodes can contain (a tree has $p - 1 = 7$ edges); the missing edge is excluded by the tree edge-count constraint, not by estimation failure.

4.10 Breast cancer gene expression (illustration only)

We apply Algorithm 1b to $n = 250$ samples and 1000 genes [24]. The top gene identified by both the t -test p -value and the k -NN mutual information against the class label is **A.202870_s.at**.

No ground-truth graph is available for this dataset. The section is included for illustration only. Algorithm 1b uses the Gaussian MI formula for gene–gene pairs. Figure 6 shows the top-50-gene forest learned by the proposed procedure on this dataset.

Table 9: Paired comparisons for the sample-size sweep.

n	KSG vs MDL discr. MI: paired t -test on F_1
75	MDL discr. preferred, $P = 0.0395$
100	n.s., $P = 0.2308$
200	n.s., $P = 0.1885$
500	n.s., $P = 0.2697$
1000	n.s., $P = 0.0967$ (KSG slightly higher mean)
1500	KSG preferred, $P = 0.0265$

Table 10: Estimator ablation on the eQTL benchmark.

Procedure	F_1	SHD
Alg. 1 (proposed) + ANOVA MI	0.833 ± 0.000	2.00 ± 0.00
Alg. 1 (proposed) + fixed-grid discr. MI	0.833 ± 0.000	2.00 ± 0.00
Alg. 1 (proposed) + k -NN (KSG) MI	0.833 ± 0.000	2.00 ± 0.00
Alg. 1a (Edwards-restr.) + k -NN (KSG) MI	0.667 ± 0.000	4.00 ± 0.00

5 Discussion

The propositions of Section 3.1 formalise the observation that Edwards’ restriction is unnecessary at the structure-selection step. Corollary 2 is the main quantitative contribution. It yields a sample complexity of $O(\sigma^2 \log p / \gamma_{\text{swap}}^2)$. Under (A2), $\gamma_{\text{swap}} = \gamma$ at the population level (because the second-best spanning tree differs from T^* by a single edge swap), and Corollary 2 is tighter than Corollary 1 by a factor of $(p - 1)^2$ in the exponent.

Empirically, on the $n = 1,500$ graph-size sweep with random mixed trees containing D - G - D Markov paths, removing the restriction yields higher F_1 at every tested graph size, with paired tests remaining significant after Benjamini–Hochberg correction (Section 4.4). On the eQTL benchmark (Section 4.3) the advantage is $F_1 = 0.833$ versus 0.667 ; the recovery is deterministic at this signal level. Against the MDL-discretisation MI estimator of Suzuki [22] (Section 4.7), the comparison is sample-size dependent. At $n = 75$, pairing Algorithm 1 with MDL discretisation yields significantly higher F_1 than pairing it with KSG ($P = 0.04$). At $n \in \{100, 200, 500, 1,000\}$ the two estimators are statistically indistinguishable ($P > 0.05$ in every case). At $n = 1,500$, pairing with KSG is significantly better ($P = 0.027$). On the fully discrete Asia Bayesian network (Section 4.9) all three procedures (Algorithm 1 with KSG, Algorithm 1a with KSG, Algorithm 1 with MDL discretisation) coincide because Edwards’ restriction is inactive on all-discrete data; at $n = 5,000$, the learned tree recovers 7 of the 8 skeleton edges, the maximum possible number for any spanning tree on 8 nodes.

Multiple testing. Sections 4.4 and 4.7 report 20 paired t -tests across the scaling, effect-size, heterogeneous, and MI-estimator sample-size sweeps. Of these, 15 have raw $P < 0.05$ and 5 have raw $P \geq 0.05$. After applying the Benjamini–Hochberg procedure to control the false discovery rate at 0.05, we find that 14 of the 15 raw-significant tests remain significant after correction (largest adjusted $P = 0.039$); the remaining raw-significant comparison (MDL versus KSG at $n = 75$, raw $P = 0.0395$) lies above the BH threshold (adjusted $P = 0.053$). The five raw non-significant comparisons remain non-significant under the correction. Multiple-testing adjustment therefore does not change the qualitative claim.

Appendix B reports an empirical variance-scale diagnostic for the KSG and mixed-pair k -NN estimators. The diagnostic returns the quantity $\sigma_{\text{emp}} = \sqrt{n \cdot \text{Var}(\hat{I}_n)}$, which is the empirical analogue of the sub-Gaussian parameter σ from Corollary 1. The value $\sigma_{\text{emp}} = 1.2$ is an upper envelope across all Gaussian-pair correlations and Gaussian-shift effect sizes tested in Appendix B; the largest cell in the underlying tables is 1.10 for the continuous–continuous KSG estimator and 0.89 for the mixed-pair k -NN estimator, and we round upward to 1.2 (approximately a 9% margin above the maximum observed value). The diagnostic is useful for interpreting the scale of fluctuations in the experiments, but it does not establish the sub-Gaussian deviation assumption used in Corollaries 1 and 2. The numerical substitution $\sigma = 1.2$ into the sample-complexity formula should therefore be read as illustrative under the assumed tail bound, not as a proven finite-sample guarantee. A formal derivation of σ as a function of explicit smoothness assumptions on the densities, in the spirit of Singh and Póczos [21], Berrett et al.

Table 11: Asia network skeleton recovery (the three procedures coincide on this benchmark).

n	F_1
500	0.803 ± 0.091
1000	0.800 ± 0.069
2500	0.827 ± 0.053
5000	0.933 (full sample, single run)

Table 12: Top five genes ranked by k -NN mutual information against the class label.

Rank	Probe	k -NN MI
1	A.202870_s_at	0.236
2	A.212949_at	0.214
3	A.209408_at	0.208
4	B.224753_at	0.190
5	B.228559_at	0.189

[2], and Han et al. [8], would convert the empirical diagnostic into a theorem.

Our bounds depend on a mutual-information gap (the tree-symmetric-difference gap γ in Corollary 1 and the per-swap gap γ_{swap} in Corollary 2, which coincide under (A2)), but we do not derive these gaps from more interpretable signal properties such as effect sizes or conditional variances. A future analysis that expressed γ_{swap} in terms of such quantities would tie the sample-complexity formula directly to features of the data-generating model.

Practical reach of the per-swap bound on the random-tree generator. An empirical proxy for γ_{swap} computed on the estimated population MI matrix of the random-mixed-tree generator (Appendix A.1) at $n = 8,000$ has median ≈ 0 nats (IQR $[0, 0.001]$ and maximum 0.0035 nats across 15 trials). At the median, (A2) effectively fails and the per-swap bound is undefined. Even at the largest γ_{swap} observed across the 15 trials, substituting $\gamma_{\text{swap}} = 0.0035$ nats into the Corollary 2 sample-complexity formula at $\sigma = 1.2$, $p = 20$, $\delta = 0.1$ yields $n \geq 138/(0.0035)^2 \approx 1.1 \times 10^7$. Corollary 2 is therefore not numerically informative on the random-mixed-tree benchmark used in Sections 4.4 and 4.7; the content of the bound is diluted by near-ties in the population MI matrix produced by that generator.

A bounded- γ_{swap} regime where Corollary 2 is numerically informative. As a controlled demonstration, we consider a three-node D_1 - G - D_2 Markov chain with strong effect: $D_1 \sim \text{Uniform}\{0, 1, 2\}$, $G = 2D_1 + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, 1)$, and D_2 obtained by binning G at its population terciles. The population MIs are obtained in closed form except for $H(G)$, which is computed by one-dimensional quadrature against the Gaussian mixture $(1/3) \sum_{d=0}^2 \mathcal{N}(2d, 1)$. The conditional entropy $H(G | D_1)$ equals $\frac{1}{2} \log(2\pi e)$, since each conditional density is $\mathcal{N}(2d, 1)$. The joint $P(D_1, D_2)$ is a sum of three normal-CDF differences, giving $H(D_1, D_2)$ in closed form. The mutual information $I(G; D_2) = \log 3$ because D_2 is a deterministic function of G at the population terciles. This gives $I(D_1; G) = 0.6191$ nats, $I(G; D_2) = 1.0986$ nats, $I(D_1; D_2) = 0.5202$ nats, and a per-swap gap $\gamma_{\text{swap}} = 0.0990$ nats. Substituting into Corollary 2 at $\sigma_{\text{emp}} = 1.2$, $p = 3$, $\delta = 0.1$ gives a required sample size of $n \approx 7,400$; an empirical evaluation over 100 trials at $n = 1,500$ achieves 100% recovery on this regime. The bound is therefore conservative but numerically finite when γ_{swap} is bounded away from zero.

Extension to $p = 15$. To verify that the same finite-sample story holds beyond the three-node case, we form a $p = 15$ generator by stacking five independent copies of the above D_1 - G - D_2 chain. The population MI matrix is block-diagonal, so the per-block $\gamma_{\text{swap}} = 0.0990$ nats from the three-node case carries over. Substituting into Corollary 2 at $\sigma = 1.2$, $p = 15$, $\delta = 0.1$ gives a required sample size of $n \approx 13,100$. The truth is a 10-edge forest, whereas the maximum-spanning-tree step returns a 14-edge spanning tree on the 15 nodes; we therefore report whether all ten true edges are included in the returned spanning tree. Across 100 trials at $n = 1,500$, all ten true edges are present in the returned tree in 100% of trials, with edge-set $F_1 = 0.833$.¹ The bound’s prediction is satisfied well before the formal required

¹Because across-block population MI is exactly zero in this construction, assumption (A2) of unique-maximum MST does not hold globally. The demonstration is therefore most cleanly viewed as an instance of Proposition 2, the thresholded-

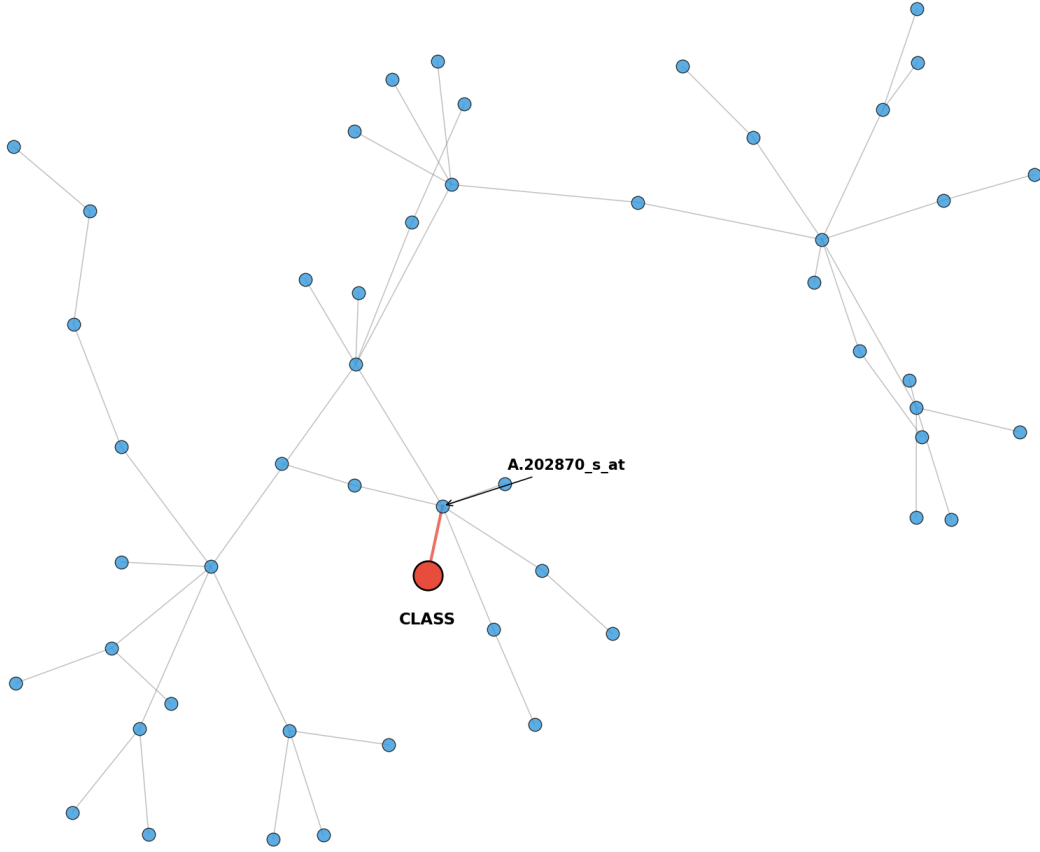


Figure 6: Chow-Liu forest learned by Algorithm 1b on the breast-cancer expression dataset (top 50 differentially expressed genes plus the binary class label, $n = 250$). Gene-class edges use the mixed-pair k -NN mutual information estimator; gene-gene edges use the Gaussian formula. The class node (CLASS, red) is connected to gene A.202870_s_at, which is also the top gene by t -test p -value. The section is presented as an illustration; no ground-truth graph is available for this dataset.

sample size, as expected of a conservative concentration bound. The general message is that Corollary 2 is informative for problems whose population MI matrix has separated edge weights, and that it becomes numerically vacuous when the population matrix admits near-ties; the random-mixed-tree generator used in Section 4.4 and Section 4.7 produces the latter regime.

6 Conclusion

We have revisited the mixed-type Chow-Liu algorithm and asked whether Edwards’ type-pair restriction, which forbids a continuous variable from connecting two discrete variables along a tree path, is needed at the structure-selection stage. We showed that it is not. Tree selection depends only on pairwise mutual-information values, each symmetric in its two arguments and well-defined for every combination of variable types, so the Chow-Liu objective does not require a directional local factor such as $P(D | G)$. The restriction enters only at the subsequent step that fits a parametric density on the chosen tree.

Proposition 1 shows that the unrestricted procedure recovers the true Chow-Liu tree as the sample size grows, whenever the pairwise mutual-information estimator is per-pair consistent. Proposition 2 extends this conclusion to a thresholded forest analogue under a shrinking-threshold rule. Corollaries 1 and 2 supply finite-sample tree-recovery bounds under a per-pair sub-Gaussian deviation assumption with a common parameter σ , and the per-swap bound in Corollary 2 sharpens the tree-symmetric-difference

forest version: under a shrinking threshold τ_n that excludes near-zero MI edges, the returned forest is exactly the ten-edge truth and edge-set F_1 tends to 1. We report the spanning-tree statistic above to keep the comparison directly aligned with the three-node demonstration in the preceding paragraph.

bound in Corollary 1 by a factor of $(p - 1)^2$ in the exponent. Appendix B reports a variance-scale diagnostic that supports the sub-Gaussian substitution numerically without proving the tail itself.

Empirically, the unrestricted procedure outperforms the Edwards-restricted baseline on random mixed graphs containing D - G - D Markov paths that the restricted procedure cannot represent. Of the 20 paired t -tests reported in Sections 4.4 and 4.7, 15 are significant at $\alpha = 0.05$ and 14 remain significant under Benjamini–Hochberg correction at the same level. The synthetic eQTL network corroborates the recovery improvement on a biologically motivated structure, and the fully discrete Asia Bayesian benchmark, where Edwards’ restriction is inactive by construction, serves as a separate sanity check that the pipeline attains the tree-approximation ceiling on a canonical discrete network. Substituting the MDL-discretisation estimator for the k -NN estimator inside the same unrestricted pipeline produces a sample-size-dependent comparison. MDL discretisation outperforms k -NN at $n = 75$; the two estimators perform indistinguishably at intermediate sample sizes ($n \in \{100, 200, 500, 1,000\}$); and k -NN outperforms MDL discretisation at $n = 1,500$.

Two caveats apply. First, the returned graph is undirected, and its edges represent statistical associations rather than causal relations. Second, the result covers structure selection only. If a later step fits a parametric density on the chosen tree, Edwards’ restriction or an alternative parametrisation of the offending local conditional becomes relevant again at that step, and we do not address it here.

Funding

No funding was received for this project.

Conflict of interest

The authors declare no conflict of interest.

Code and data availability

The full reproduction code, including the random-mixed-tree generator, the experiment scripts, and the figure-generation pipeline, is available at the repository

<https://github.com/ash141886/unrestricted-chow-liu->

The Asia network dataset is included with the `gRbase` R package [9], and the breast-cancer gene-expression dataset of West et al. [24] is publicly available.

Declaration on the use of generative AI

During the preparation of this manuscript, the authors used a large language model (ChatGPT) to assist with writing polish, notation consistency checks, and LaTeX formatting. All theoretical results, derivations, experimental designs, software implementations, and final scientific claims were originated and verified by the authors. The authors reviewed and edited the AI-assisted content and take full responsibility for the content of the publication.

Appendix

The appendix contains the proofs of Propositions 1 and 2, the proofs of Corollaries 1 and 2, the data-generation rules used by the random-mixed-tree generator, the empirical variance-scale diagnostic supporting the sub-Gaussian substitutions used in Corollaries 1 and 2, and an exploratory adaptive- k table.

A Proofs and finite-sample bounds

Proof of Proposition 1

Define the population objective $W(T) = \sum_{(i,j) \in T} I(X_i; X_j)$ and its sample analogue $\hat{W}_n(T) = \sum_{(i,j) \in T} \hat{I}_n(X_i; X_j)$. The Chow–Liu estimator is $\hat{T}_n = \arg \max_{T \in \mathcal{T}} \hat{W}_n(T)$, and the target is $T^* = \arg \max_{T \in \mathcal{T}} W(T)$.

Step 1 (uniform convergence over $\mathcal{T}$). Every spanning tree has exactly $p - 1$ edges, so

$$\sup_{T \in \mathcal{T}} |\hat{W}_n(T) - W(T)| \leq (p - 1) \max_{i < j} |\hat{I}_n(X_i; X_j) - I(X_i; X_j)| \xrightarrow{p} 0,$$

where the convergence follows from (A3) applied to each of the $\binom{p}{2}$ pairs.

Step 2 (identification). By (A2), T^* is unique, so there exists a positive gap $\gamma = W(T^*) - \max_{T \neq T^*} W(T) > 0$.

Step 3 (consistency). For any $T \neq T^*$,

$$\hat{W}_n(T^*) - \hat{W}_n(T) \geq \gamma - 2 \sup_{T'} |\hat{W}_n(T') - W(T')|.$$

The right-hand side exceeds $\gamma/2 > 0$ with probability tending to 1, so $P(\hat{T}_n = T^*) \rightarrow 1$. This is the standard M-estimation argument [23]. $\square$

Remark (no local CG parametrisation $P(D | G)$ is required). For a mixed pair (G, D) , mutual information is symmetric:

$$I(G; D) = H(G) - H(G | D) = H(D) - H(D | G). \quad (2)$$

The Chow–Liu objective depends only on the scalar pairwise quantity $I(G; D)$, not on a chosen directed local factorisation of the pair. Evaluating or estimating an edge weight therefore does not require specifying a standard conditional-Gaussian local factor $P(D | G)$. Such a factor may be required at a later, separate step that writes the joint distribution as an explicit parametric density on the selected tree.

A.1 Data-generation rules for the random-mixed-tree generator

The random-mixed-tree generator uses a Prüfer sequence to sample a labelled tree uniformly. Conditional on the tree, data are generated in BFS order from a randomly chosen root, using the following parent-to-child rules:

- Discrete parent D , continuous child G : $G = \text{effect} \cdot D + \varepsilon$, with $\varepsilon \sim \mathcal{N}(0, 1)$.
- Continuous parent G , discrete child D : $D = \mathbf{1}[G > q_{1/3}] + \mathbf{1}[G > q_{2/3}]$, where $q_{1/3}$ and $q_{2/3}$ are the empirical tercile cutoffs of G .
- Two discrete: noisy copy with a 30% flip rate.
- Two continuous: $G_{\text{child}} = \alpha G_{\text{parent}} + 0.8\varepsilon$, with $\alpha = \text{effect}/(1 + \text{sd}(G_{\text{parent}}))$.

A.2 Tree-symmetric-difference bound

Corollary 1 (Tree-symmetric-difference bound). *Assume (A1)–(A2) and that, for every pair (i, j) with a common parameter $\sigma > 0$, the pairwise estimator satisfies the sub-Gaussian tail*

$$P(|\hat{I}_n(X_i; X_j) - I(X_i; X_j)| > t) \leq 2 \exp\left(-\frac{nt^2}{2\sigma^2}\right), \quad t > 0. \quad (3)$$

This tail implies (A3) and so subsumes the consistency hypothesis of Proposition 1. With $\gamma = W(T^) - \max_{T \neq T^*} W(T) > 0$,*

$$P(\hat{T}_n \neq T^*) \leq p^2 \exp\left(-\frac{n\gamma^2}{8\sigma^2(p-1)^2}\right). \quad (4)$$

Proof. From the uniform-bound step of Proposition 1 (Step 1), $\hat{T}_n = T^*$ whenever $\sup_{T \in \mathcal{T}} |\hat{W}_n(T) - W(T)| < \gamma/2$, which in turn holds whenever $\max_{i < j} |\hat{I}_n(X_i; X_j) - I(X_i; X_j)| < \gamma/(2(p-1))$. Hence

$$P(\hat{T}_n \neq T^*) \leq P\left(\max_{i < j} |\hat{I}_n(X_i; X_j) - I(X_i; X_j)| \geq \gamma/(2(p-1))\right).$$

Applying a union bound over the $\binom{p}{2}$ pairs combined with the sub-Gaussian tail (3) evaluated at $t = \gamma/(2(p-1))$, and using $2\binom{p}{2} = p(p-1) \leq p^2$, we obtain the stated bound. Setting the right-hand side equal to δ and solving for n yields

$$n \geq \frac{8\sigma^2(p-1)^2(2\log p + \log(1/\delta))}{\gamma^2}.$$

$\square$

A.3 Per-swap bound

For each non-tree edge $e' \notin T^*$, let $\text{Cyc}_{T^*}(e')$ denote the fundamental cycle of e' relative to T^* , defined as the unique cycle obtained by adding e' to T^* . Optimality of Kruskal's algorithm on the population implies that for every $e' \notin T^*$ and every $e \in \text{Cyc}_{T^*}(e') \cap T^*$, $I(e) \geq I(e')$. Under (A2) the inequality is strict. We define the per-swap mutual information gap as

$$\gamma_{\text{swap}} = \min_{e' \notin T^*} \min_{e \in \text{Cyc}_{T^*}(e') \cap T^*} [I(e) - I(e')] > 0. \quad (5)$$

Corollary 2 (Per-swap finite-sample bound). *Under (A1)–(A2) and the per-pair sub-Gaussian tail (3) with common parameter $\sigma > 0$,*

$$P(\hat{T}_n \neq T^*) \leq 2p^3 \exp\left(-\frac{n\gamma_{\text{swap}}^2}{8\sigma^2}\right). \quad (6)$$

The associated sample complexity is $n \geq 8\sigma^2(3\log p + \log(2/\delta))/\gamma_{\text{swap}}^2 = O(\sigma^2 \log p / \gamma_{\text{swap}}^2)$.

Proof. By the matroid optimality characterisation of maximum spanning trees, if $\hat{I}_n(e) > \hat{I}_n(e')$ for every non-tree edge $e' \notin T^*$ and every tree edge $e \in \text{Cyc}_{T^*}(e') \cap T^*$, then T^* is the unique maximum spanning tree under $\hat{I}_n$ and Kruskal's algorithm returns $\hat{T}_n = T^*$. Taking the contrapositive, $\hat{T}_n \neq T^*$ implies that for at least one such pair (e, e') , $\hat{I}_n(e) \leq \hat{I}_n(e')$.

For a single pair (e, e') with population gap $\gamma_{e,e'} = I(e) - I(e') > 0$,

$$\begin{aligned} P(\hat{I}_n(e) \leq \hat{I}_n(e')) &\leq P(|\hat{I}_n(e) - I(e)| \geq \gamma_{e,e'}/2) + P(|\hat{I}_n(e') - I(e')| \geq \gamma_{e,e'}/2) \\ &\leq 4 \exp\left(-\frac{n\gamma_{e,e'}^2}{8\sigma^2}\right). \end{aligned}$$

Each non-tree edge e' has a fundamental cycle of length at most p , containing at most $p-1$ tree edges (the cycle consists of e' together with the unique tree path between its endpoints). The number of non-tree edges is at most $\binom{p}{2} - (p-1) \leq p^2/2$. A union bound over all relevant pairs, using $\gamma_{e,e'} \geq \gamma_{\text{swap}}$, yields

$$P(\hat{T}_n \neq T^*) \leq \frac{p^2}{2} \cdot (p-1) \cdot 4 \exp\left(-\frac{n\gamma_{\text{swap}}^2}{8\sigma^2}\right) \leq 2p^3 \exp\left(-\frac{n\gamma_{\text{swap}}^2}{8\sigma^2}\right).$$

□

Remark. For Kruskal's algorithm under (A2) uniqueness, the second-best spanning tree differs from T^* by a single edge swap, so the global gap γ and the per-swap gap γ_{swap} coincide at the population level. Corollary 2 is then tighter than Corollary 1 by a factor of $(p-1)^2$ in the exponent. The bound is numerically informative for problems with γ_{swap} bounded away from zero; we discuss the practical reach of the bound on our random-mixed-tree generator in Section 5 (Limitations).

B Empirical variance-scale diagnostic

The bounds in Corollaries 1 and 2 are conditional finite-sample bounds: they hold whenever the pair-wise mutual information estimator satisfies the sub-Gaussian deviation inequality $P(|\hat{I}_n - I| > t) \leq 2 \exp(-nt^2/(2\sigma^2))$ with the stated σ . Finite-sample sub-Gaussian constants for the KSG and mixed-pair k -NN estimators have not, to our knowledge, been established in the literature. We do not prove the sub-Gaussian condition. Instead, we report (i) a worst-case McDiarmid-type calculation, (ii) an empirical variance-scale diagnostic, and (iii) a multi-quantile tail check on the diagnostic. The latter two are empirical and do not constitute a proof of sub-Gaussian behaviour. The discrete-discrete plug-in estimator (used on Asia) is bounded above by $\log \min(K_i, K_j)$ and so satisfies a sub-Gaussian tail by Hoeffding's inequality with a constant depending on the support sizes; we do not separately calibrate this case.

Worst-case theoretical bound (McDiarmid). Each summand in the KSG estimator is $\psi(n_x(i) + 1) + \psi(n_y(i) + 1)$, where ψ is the digamma function. The summand is bounded above by $\psi(N) \leq \log(N + 1)$. Changing one data point changes the k -nearest-neighbour structure of at most $O(k)$ other points, so the bounded-difference constants are $c_i = O((k \log N)/N)$. McDiarmid’s inequality $P(|\hat{I}_n - \mathbb{E}\hat{I}_n| > t) \leq 2 \exp(-2t^2 / \sum_i c_i^2)$ then gives a sub-Gaussian tail with $\sigma^2 = O((k \log N)^2/4)$. For $k = 5$ and $N = 1,500$, this gives $\sigma \approx k \log(N)/2 \approx 18$ nats, which is too loose to be informative in practice.

Empirical variance-scale diagnostic. The McDiarmid bound above is a worst-case constant derived from the bounded-difference property of the estimator and is too loose to be useful in practice. A practical alternative is to estimate the asymptotic variance scale directly. The asymptotic-variance relation $\text{Var}(\hat{I}_n) \approx \sigma^2/n$ suggests an empirical diagnostic. We draw $\hat{I}_n$ from 200 independent samples of size n , compute the sample variance, and report $\sigma_{\text{emp}} = \sqrt{n \cdot \text{Var}(\hat{I}_n)}$. We caution that this diagnostic estimates a variance scale; a bounded variance does not, on its own, imply a sub-Gaussian tail (a distribution can have bounded variance and polynomial rather than Gaussian tails). The reported σ_{emp} is therefore the σ that one would substitute if the sub-Gaussian deviation inequality held, not an established constant. The procedure is repeated for several sample sizes $n \in \{200, 500, 1,500, 5,000\}$ to verify the $n^{-1/2}$ scaling.

Table 13: Empirical σ for the KSG continuous–continuous estimator on Gaussian pairs.

ρ	True MI	Median σ_{emp}
0.00	0.0000	0.34
0.30	0.0472	0.63
0.50	0.1438	0.77
0.70	0.3367	0.94
0.90	0.8304	1.10

Table 14: Empirical σ for the mixed-pair k -NN estimator on Gaussian-shift data with $K = 3$ discrete categories.

effect	True MI	Median σ_{emp}
0.5	0.0771	0.76
1.0	0.2538	0.89
1.5	0.4472	0.85
2.0	0.6191	0.82

The empirical σ is approximately constant in n , which confirms the $n^{-1/2}$ scaling of $\sqrt{\text{Var}(\hat{I}_n)}$. It increases with the underlying MI for the continuous–continuous estimator and is approximately constant in effect size for the mixed-pair estimator. Across all settings tested, $\sigma_{\text{emp}} \leq 1.2$. Illustratively substituting the empirical variance-scale value $\sigma_{\text{emp}} = 1.2$ into the Corollary 2 sample-complexity formula gives

$$n \geq \frac{8 \cdot 1.2^2 (3 \log p + \log(2/\delta))}{\gamma_{\text{swap}}^2} = \frac{11.52 (3 \log p + \log(2/\delta))}{\gamma_{\text{swap}}^2}. \quad (7)$$

For $p = 20$ and $\delta = 0.1$, this evaluates to $n \geq 138/\gamma_{\text{swap}}^2$. At $\gamma_{\text{swap}} = 0.1$ nats, $n \geq 13,800$; at $\gamma_{\text{swap}} = 0.3$ nats, $n \geq 1,530$, which is comparable to the $n = 1,500$ used in the experiments.

Multi-quantile tail check. The variance-scale estimate above does not test the tail behaviour. As a falsifiable diagnostic, we draw 2000 estimates of $\hat{I}_n$ on Gaussian pairs at $\rho = 0.5$ (true MI ≈ 0.144) and $n = 1,500$, and compare the empirical exceedance probabilities $\hat{P}(|\hat{I}_n - I| > t)$ to the Gaussian bound $2 \exp(-nt^2/(2\sigma_{\text{emp}}^2))$ at several t values. For all tested t , the empirical exceedance probabilities are bounded above by the corresponding Gaussian bound (Table 15).

At every informative t , the empirical exceedance probability is bounded above by the Gaussian bound. This is consistent with sub-Gaussian behaviour for this estimator on this distribution, but it does not prove sub-Gaussian behaviour in general.

Table 15: Multi-quantile diagnostic for the KSG estimator at $\rho = 0.5$, $n = 1,500$, 2000 trials. Empirical exceedance probability vs. the Gaussian bound at the empirically estimated variance scale $\sigma_{\text{emp}} = 0.74$. Rows with bound > 1 are vacuous (probabilities are bounded by 1 by definition) and are omitted; the informative range begins around $t = 0.03$.

t	$\hat{P}(\hat{I}_n - I > t)$	$2 \exp(-nt^2/(2\sigma_{\text{emp}}^2))$
0.030	0.110	0.590
0.040	0.038	0.228
0.050	0.009	0.067
0.100	0.000	0.000
0.200	0.000	0.000

Scope of the diagnostic. The empirical σ_{emp} is computed on Gaussian-pair and Gaussian-shift conditional families only. We do not test the bimodal, skewed, or heavy-tailed conditional families used in Section 4.1; the calibration is therefore narrower than the experiments in Section 4.

Rigorous finite-sample concentration for the KSG estimator under explicit smoothness assumptions has been studied in related directions by Singh and Póczos [21] for k -NN entropy estimators, Berrett et al. [2] for efficient k -NN entropy under analytic densities, and Han et al. [8] for nonparametric divergence estimators. A formal derivation of σ as a function of density-smoothness parameters along these lines would convert our empirical diagnostic into a theorem.

C Exploratory adaptive- k table

As an exploratory check using 10 trials per sample size, we test the rule $k(n) = \max(5, \lceil 150/\sqrt{n} \rceil)$ on the same MI-estimator sample-size sweep used in Section 4.7.

Table 16: Adaptive- k variant against the MDL-discretisation MI estimator [22], $p = 15$, effect = 0.6, 10 trials per row (exploratory).

n	$k(n)$	F_1 adaptive- k	F_1 MDL discr.	paired P
75	18	0.471 ± 0.097	0.566 ± 0.109	0.0068
100	15	0.471 ± 0.125	0.526 ± 0.112	0.2163
200	11	0.557 ± 0.149	0.619 ± 0.112	0.0711
500	7	0.664 ± 0.120	0.671 ± 0.107	0.8760
1000	5	0.679 ± 0.140	0.629 ± 0.119	0.2259
1500	5	0.664 ± 0.079	0.664 ± 0.096	1.0000

The adaptive rule narrows but does not eliminate the small-sample advantage of the MDL-discretisation MI estimator. The trial count is smaller than that used in the main MI-estimator comparison (Table 8), so the adaptive- k check is presented as exploratory only and is not used to revise the main $n = 1,500$ comparison.

D Proof of Proposition 2

We partition the $\binom{p}{2}$ pairs into $\mathcal{E}_+ = \{(i, j) \in F^*\}$, $\mathcal{E}_0 = \{(i, j) : i, j \text{ lie in distinct components of } F^*\}$, and $\mathcal{E}_c$ (the remaining pairs). By (A4), $I \geq \delta > 0$ on $\mathcal{E}_+$ and $I = 0$ on $\mathcal{E}_0$.

Step 1. For $(i, j) \in \mathcal{E}_+$, (A3) together with $\tau_n \rightarrow 0$ implies that for sufficiently large n , $\tau_n < \delta/2$, so $P(\hat{I}_n > \tau_n) \rightarrow 1$.

Step 2. For $(i, j) \in \mathcal{E}_0$, (A5) gives $P(|\hat{I}_n| > \tau_n) \rightarrow 0$, so $P(\hat{I}_n \leq \tau_n) \rightarrow 1$. Note that (A5) is strictly stronger than (A3). The condition (A3) only gives pointwise convergence $\hat{I}_n \xrightarrow{p} 0$ at any fixed threshold, but here τ_n itself shrinks to zero; (A5) is the joint requirement that the estimator's convergence rate dominates the shrinkage rate of τ_n . The accompanying condition $\tau_n \sqrt{n} \rightarrow \infty$ ensures that τ_n dominates the $O_p(n^{-1/2})$ fluctuations of the estimator.

Step 3. Pairs in $\mathcal{E}_c$ have both endpoints in the same component of F^* . On the event from Step 2, where every cross-component edge in $\mathcal{E}_0$ has been removed by the threshold rule, Kruskal's algorithm

reduces to an independent maximum-spanning-tree problem on each connected component of F^* , because no surviving candidate edge crosses components.

Fix one component C and consider this restricted problem on C :

- the vertex set is C ;
- the candidate edges are $\{(i, j) \in \mathcal{E}_+ \cup \mathcal{E}_c : i, j \in C\}$;
- the population edge weights are the pairwise mutual informations $I(X_i; X_j)$ for $i, j \in C$;
- the population target is the maximum spanning tree on C under these weights, which by (A2') is unique and equals the restriction of F^* to C .

The hypotheses of Proposition 1 hold on this restricted problem: (A1) is invariant under restriction, (A3) is a pairwise condition that still holds for every pair in C , and (A2') gives uniqueness of the target tree on C . Applying Proposition 1 to the restricted problem therefore yields that Kruskal's algorithm run on the within-component candidate edges returns the restriction of F^* to C with probability tending to 1.

Repeating this argument for every component of F^* gives that the empirical forest matches F^* on $\mathcal{E}_+$ with probability tending to 1. Any further edge from $\mathcal{E}_c$ considered later by Kruskal would close a cycle within an already-built component, and is therefore rejected.

Step 4. The intersection of the events in Steps 1 to 3 has probability tending to 1. On this event, Kruskal's algorithm with threshold τ_n returns exactly F^* by (A2'). $\square$

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